10.7.6 Wrap-Up: Cool Down

1. Write a short note to a classmate who missed class summarizing today's lesson.

Unit 10, Lesson 8: Random Samples vs. Population

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

- ☐ I can determine if a sample is representative of the population.

10.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy is an electrical vehicle technician for City Car Share. When Ben was considering what to do with his life, he knew he wanted to do something that would bring him happiness. Now he uses the skills he has gained from his study of landscape architecture and work as an electrical engineer to maintain the fleet of electric vehicles and electric bikes used by City Car Share.

Ben found that dedicating himself to a cause that is bigger than himself brings him happiness.

1. Think about what you would like to do with your life. What kind of work would make you happy?

In talking about his work, Ben said, "That's how engineering and design is. You never get to a point where you think, 'Oh, that's perfect! That's done.' You start asking questions like, 'Now that we've gotten this far, what's the next step? How can we improve this?'"

2. How do you think having this mentality is helpful?

10.8.2 Guided Instruction: Is the Sample Representative of the Population?

Many organizations want to know how large groups of people think and behave. This helps businesses make decisions regarding what to sell, when to be open, and whom to target with advertisements. Research is also used to make predictions about events like elections, and to help doctors provide appropriate healthcare to their patients.

To gather this information, companies and organizations often collect data ***samples*** from a ***population*** they are studying.

A ***population*** is the entire set of cases or individuals under consideration in a statistical analysis.

A ***sample*** is a subset, or part, of a population.

1. Imagine a group of researchers want to gather information about which candidate for sheriff all registered voters in Glades County, Florida plan to vote for in an upcoming election.

   Use the definitions to complete the statements.

   In this scenario, the [○ population ○ sample] being studied is all registered voters in Glades County, Florida. It is not realistic for researchers to ask every single registered voter, so researchers would collect data from a [○ population ○ sample] of registered voters in Glades County, Florida to make predictions.

2. Volusia County School District wants to find out more about which mode of transportation students take to school, so they decide to conduct a survey. The entire school district has a population of $63,223$ students. Since the number of students is so large, the district will only survey some of the students.

   a. In this scenario, what is the population being studied?

   b. What is the sample being used?

To ensure the students surveyed are representative of all students in Volusia County, a system is used to select who is surveyed.

> A sample that is **representative** of a population resembles the population and can be used to make accurate predictions about the population.

3. Twenty-five students from each grade level will be chosen at random to participate in the survey.
   With your partner, discuss how you think this method will help ensure the sample is representative of the population.

4. Claire wants to collect data on the highest eduation level achieved by adults in Jacksonville, Florida. She decides to go to the University of North Florida campus and ask the first person who comes into the library every $\mathbf{30}$ minutes about their education level.

   a. What is the population?

   b. What is the sample?

   c. Explain why a sample chosen in this way may **not** be representative of the population.

> A ***random sampling*** is a smaller group of people or objects chosen from a larger group or population by a process giving equal chance of selection to all possible people or objects, and all possible subsets of the same size.

10.8.3 Collaboration: Comparing Methods for Selecting Samples

For each scenario below, read the information about the study and answer the questions that follow.

Scenario 1: Lin is running for president of the seventh grade. She wants to predict her chances of winning the election. She is considering three different methods for surveying a sample of the students.

1. Complete the table to describe the benefits and problems with each of the possible survey methods described.

Sample Surveyed	Benefits of this Method	Problems with this Method
Method 1: Ask everyone on her basketball team who they are voting for.		
Method 2: Ask every third girl waiting in the lunch line who they are voting for.		
Method 3: Ask the first 15 seventh graders to arrive at school one morning who they are voting for.		

2. How could the different methods for selecting a sample in Scenario 1 lead to different conclusions about the population?

3. Explain which of the methods listed in the table would be the most likely to produce a sample that is representative of the population.

4. With your partner, determine a better way to select a sample for this study.

Scenario 2: Maneet, a nutritionist, wants to collect data on how much caffeine the average American drinks per day. She works with three of her colleagues to decide how to survey a sample of the population.

Method 1	Method 2
Maneet and Diarmaid believe the best way to survey is to ask the first 20 adults who arrive at a grocery store after $10{:}00$ A.M. about the average amount of caffeine they consume each day.	Faith and Santiago would rather ask the first adult who comes into a coffee shop every 30 minutes about the average amount of caffeine they consume each day.

5. What recommendation(s) would you give to this committee about how to select the most representative sample?

10.8.4 Exploration: Sample Size

Isabella is playing a computer game. Her challenge is to create a replica of a hidden spinner that she cannot see. Her score is based on two things: the accuracy of the spinner she creates and the number of spins she uses to create the spinner (the fewer spins she uses equals a higher score).

Isabella creates her first replica after 12 spins of the hidden spinner. Below is a table showing the outcomes of the first 12 spins.

Color	Number of Spins
red	2
purple	4
blue	3
yellow	3

1. Use the data from the first 12 spins to draw a sketch of what you think the hidden spinner might look like.

Isabella's spinner was not accurate, so she had the computer generate additional data. The table below shows the data from the first 36 spins of the spinner, including the 12 shown in the previous table.

Color	Number of Spins
red	3
purple	10
blue	8
yellow	4
orange	2
green	9

2. Revise your spinner based on this new data.

3. Explain which spinner you think is closest to what the hidden spinner looks like.

4. The second spinner Isabella drew was far more accurate than the first. Having additional spins gave a more accurate representation of what the spinner actually looked like. What can this simulation teach us about choosing the size of the sample in a population study?

10.8.5 Your Turn!

1. The meat department manager at a grocery store is worried some of the packages of ground beef labeled as having one pound of meat may be under-filled. He decides to take a sample of 5 packages from a shipment containing 100 packages of ground beef. The packages were numbered as they were put in the box, so each one has a different number between 1 and 100.

   Describe how the manager can select a fair sample of 5 packages.

2. Diego wants to survey a sample of students at his school to learn about the percentage of students who are satisfied with the food in the cafeteria. He decides to go to the cafeteria on a Monday and ask the first 25 students who purchase a lunch at the cafeteria if they are satisfied with the food.
   Explain whether you think this is a good way for Diego to select his sample.

3. Eleanor wants to make the most accurate prediction about who will win the student council election. Explain whether it would be better for her to conduct a random sample of 10 people or 50 people.

10.8.6 Wrap-Up: Cool Down

Anthony is conducting research to determine how students at his school get to school each day (biking, walking, car, bus, etc.). Because there are many students in his school, he will be conducting a survey of a random sample of students to make predictions about the overall population.

1. Select all the options that would produce a random sample to represent the overall population of the school.

   - ☐ Anthony will get to school before everyone else and survey the first 50 people who walk in the doors.
   - ☐ Anthony will survey everyone at his jazz band practice.
   - ☐ Anthony will survey two randomly chosen students in every homeroom.
   - ☐ Anthony will assign each student in the school a number, randomly choose 50 numbers, and survey those students.
   - ☐ Anthony will survey every third student he sees on his walk to school.

Unit 10, Lesson 9: Making Population Predictions – Part 1

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

- ☐ I can use proportional relationships to make predictions about a population.